Group A

Assignment No: 10

Title of the Assignment: Download the Iris flower dataset or any other dataset into a DataFrame.(e.g.,https://archive.ics.uci.edu/ml/datasets/Iris). Scan the dataset and give the inference as:

1. List down the features and their types (e.g., numeric, nominal) available in the dataset.
2. Create a histogram for each feature in the dataset to illustrate the feature distributions.
3. Create a boxplot for each feature in the dataset.
4. Compare distributions and identify outliers.

Download the Iris flower dataset or any other dataset into a DataFrame. (e.g., https://archive.ics.uci.edu/ml/datasets/Iris (https://archive.ics.uci.edu/ml/datasets/Iris)).

In [1]:

```
import pandas as pd
import numpy as np
csv_url = 'https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data'
col_names = ['Sepal_Length','Sepal_Width','Petal_Length','Petal_Width','Species']
```

In [2]:

```
iris = pd.read_csv(csv_url, names = col_names)
```

Q1. How many features are there and what are their types?

In [3]:

```
column = len(list(iris))
column
```

Out[3]:

5

Clearly, dataset has 5 column indicating 5 features about the data

In [4]:

```
iris.info()
```

<class 'pandas.core.frame.DataFrame'>RangeIndex:
150 entries, 0 to 149 Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	Sepal_Length	150 non-null	float64
1	Sepal_Width	150 non-null	float64
2	Petal_Length	150 non-null	float64
3	Petal_Width	150 non-null	float64
4	Species	150 non-null	object

dtypes: float64(4), object(1)memory usage:

6.0+ KB

Hence the dataset contains 4 numerical columns and 1 object column

In [6]:

```
np.unique(iris["Species"])
```

Out[6]:

array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)

Q2. Compute and display summary statistics for each feature available in the dataset.

In [13]:

```
iris.describe()
```

Out[13]:

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

Q3. Data Visualization-Create a histogram for each feature in the dataset to illustrate the feature distributions Plot each histogram

Solution 1:

In [8]:

```
import seaborn as sns
import matplotlib
import matplotlib.pyplot as plt
%matplotlib inline
```

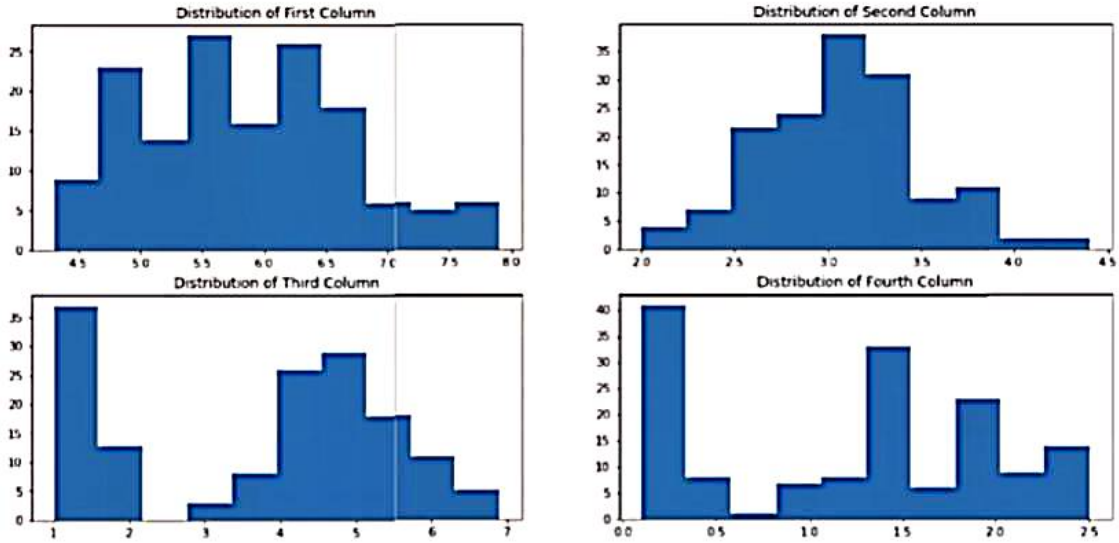
```
fig, axes = plt.subplots(2, 2, figsize = (16, 8))


axes[0,0].set_title("Distribution of First Column")
axes[0,0].hist(iris["Sepal_Length"]);

axes[0,1].set_title("Distribution of Second Column")
axes[0,1].hist(iris["Sepal_Width"]);

axes[1,0].set_title("Distribution of Third Column")
axes[1,0].hist(iris["Petal_Length"]);

axes[1,1].set_title("Distribution of Fourth Column")
axes[1,1].hist(iris["Petal_Width"]);
```

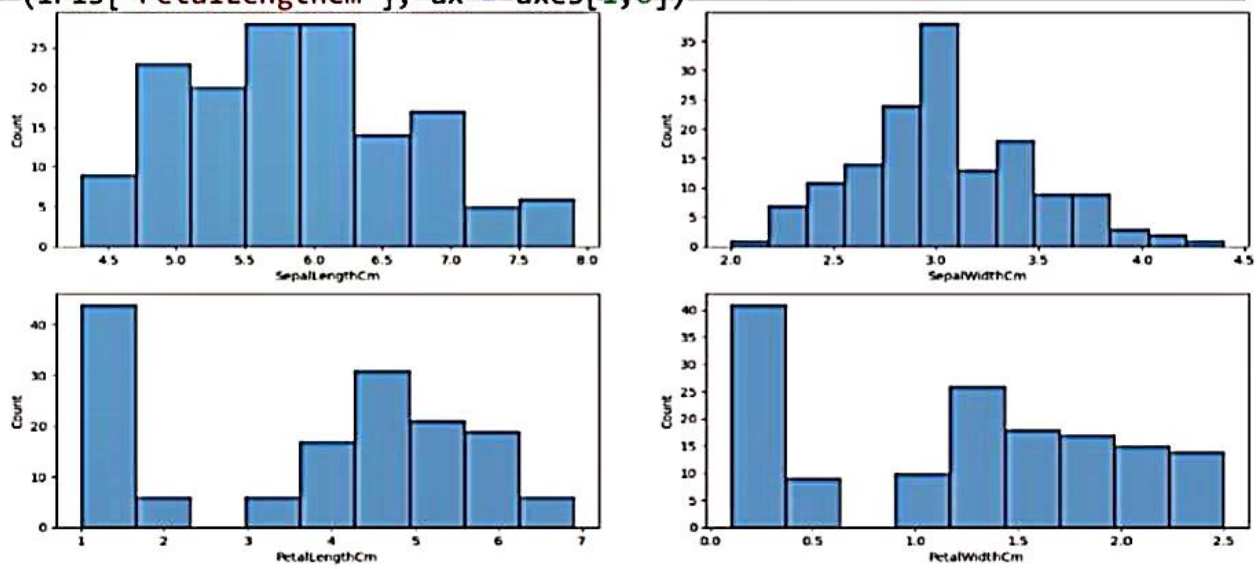


OR

Solution 2:

```
import matplotlib.pyplot as plt

fig, axes = plt.subplots(2, 2, figsize = (16,
8)) sns.histplot (iris['SepalLengthCm'], ax =
axes[0,0])sns.histplot (iris['SepalWidthCm'],
ax = axes[0,1]) sns.histplot
(iris['PetalLengthCm'], ax = axes[1,0])
```



Q4. Create a boxplot for each feature in the dataset. All of the boxplots should be combined into a singleplot. Compare distributions and identify outliers.

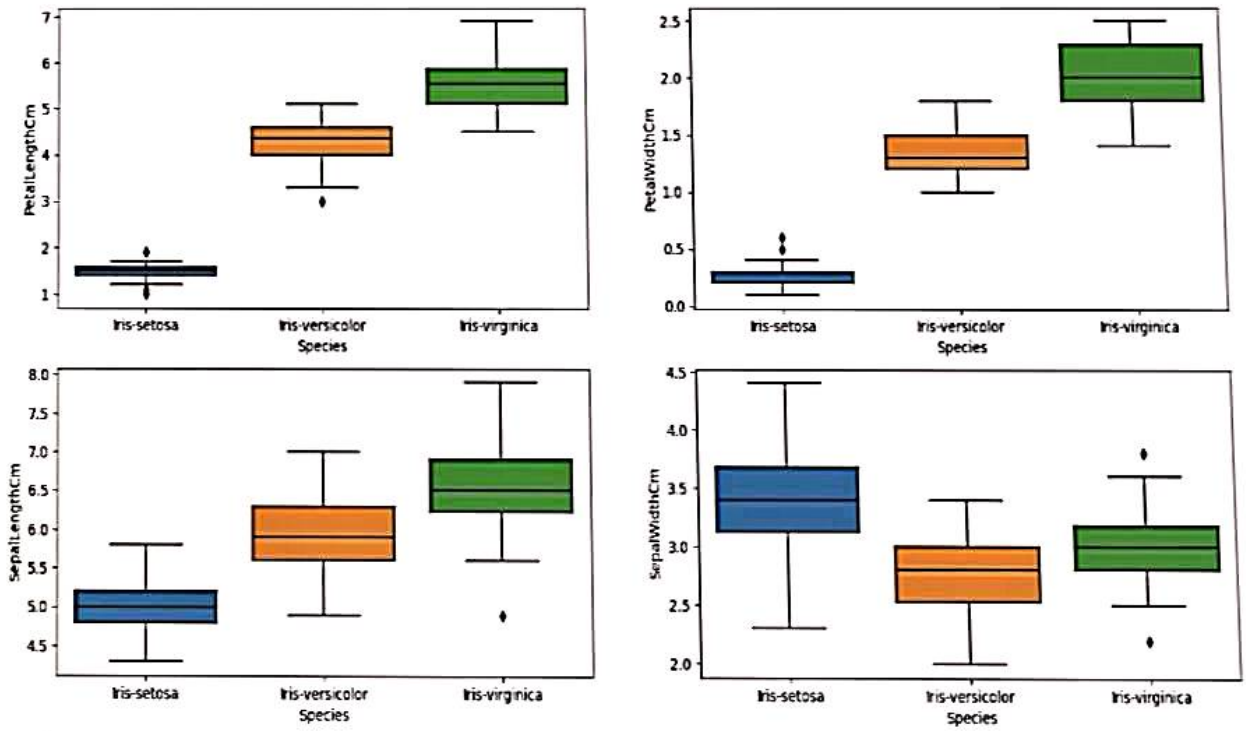
In [12]:

```
import matplotlib.pyplot as plt

fig, axes = plt.subplots(2, 2, figsize = (16, 8))
sns.boxplot(y='PetalLengthCm', x='Species', data = iris,ax =
axes[0,0])sns.boxplot(y='PetalWidthCm', x='Species', data =
iris,ax = axes[0,1]) sns.boxplot(y='SepalLengthCm', x='Species',
data = iris,ax = axes[1,0])sns.boxplot(y='SepalWidthCm',
```

Out[8]:

<AxesSubplot:xlabel='Species', ylabel=SepalWidthCm'>



If we observe closely for the box 2, interquartile distance is roughly around 0.75 hence the values lying beyond this range of (third quartile + interquartile distance) i.e. roughly around 4.05 will be considered as outliers.

Similarly outliers with other boxplots can be found.

Conclusion-

Seaborn is an advanced data visualisation library built on top of Matplotlib library. In this assignment, we looked at how we can draw histogram and boxplot for each feature in the dataset to illustrate the feature distributions